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Design of Decision Support System Framework for Course Selection

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It is evident from a review of the pertinent literature that choosing the right course frequently caused difficulties for both parents and students. There is a shortage of research that focuses on choosing among several graduating courses while taking into account all demographic, personal, academic, and social elements. Studies that focus on students' academic achievement and placement are few. In order to fill this gap, authors used the feature selection technique mutual information gain on the obtained data to identify the contributing factors and applied random forest classifier to developed decision support system framework. The current study employed a survey-based descriptive research methodology. Online surveys were utilized to gather information from students enrolled in courses in a variety of fields, including medicine, engineering, animation, cyber security, commerce and management. To choose study participants, the convenience sampling method was used. For the study, 138 students' data from all streams were gathered. The study made DSS using flask .application executed on Google Colab using ngrok. From the result it is showed that random forest algorithm performed well with 89% accuracy.

Keywords: Influencing Factors, Classifier, Feature Selection, Decision Support System.

Introduction

A decision support system (DSS) is a technology and application that can assist decision makers in gathering relevant raw data and valuable information to recognize and address issues. The user interface, appropriate data, model, and information gained from modeling data make up the majority of the DSS [1]. An information system at the management level of an organization known as a "decision support system" combines data and advanced analytical models to support decision-making in semi-structured and unstructured situations [2].

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The impact of rising population expansion has boosted demand for higher-quality education as well. Any nation must urgently raise the standards and procedures for its educational and research systems and practices if it is to develop [3]. Students' choices of courses set off a series of events that have an impact on subsequent course selections, skill development, and employment decisions. If this problem is not taken into account, wasteful resource use and subsequent student unhappiness may result because they deliver relevant and individualized information in the interest of students, recommender systems for learning tools are becoming more and more important. Artificial intelligence approaches could be helpful in this regard because they can create and mimic human decision-making, as well as using logic to reduce uncertainty to promote successful learning and a system for lifelong learning [4]. In order to make decisions in the field of academic planning, a significant amount of analysis of huge data volumes coming from numerous systems is required [5]. Numerous industries, including financial analysis, the telecommunications industry, the retail industry, the health care industry, etc. have shown the value of data mining techniques. They are crucial to the process of decision-making. One of these fields is online education, where it is essential to monitor and forecast students' performance statistics [6]. Numerous factors play a role in how students choose their academic programme [7]. Many categorization methods, including K-nearest-neighbor, Random Forest, XGBoost, and Neural Networks, have also been proposed to categories diverse types of data. Two popular ensemble concepts are bagging and boosting. Bagging combines the results from several models, whereas boosting creates a stronger model from several weaker models [8].

Following are the research objectives:

1) To select various influencing factors while course selection.
2) To create a decision support system framework for course suggestions after the 12th grade.

The remaining section of the paper is structured as follows: Section II discusses the literature; Section III explains the methodology and Section IV highlights the results and discussion followed by Section V conclusion and future enhancements.

Review of Literature

This section discusses various techniques used in the literature to tackle above mentioned challenges. Section A describes and focus on factors, techniques and area where consisted for the study, section B elaborates research gap.

Section A:

Developed Decision support system to assist users in selecting colleges and to aid in decision-making. The control, data storage, data manipulation, and model building modules make up the CS-DSS. The control module initially

directs the user through the system. Second, the data storage module houses a relational database with all the data related to collages. Third, the dBase III programming language is used to write the data manipulation module. It presents all the college information in a user-friendly way, making it possible for non-computer users to gather the information they require. The model creation module, which uses point ratings and a multi-criteria decision-making process, is lastly developed in Microsoft Quick BASIC. High school students might use this DSS as a useful tool when choosing colleges [9]. In [10] investigating the many methods of student performance data analysis and presentation for academic decision-making, the Performance-based Academic Decision-Support System (PADSS) software programme is created. The selected performance metrics can be seen at the level of each student, department, school, and university. Showed the value of an academic decision-support system (ADSS) in analyzing massive volumes of information on students' courses. a recommender system for e-Learning course suggestion is proposed using the collaborative filtering process. The study mostly concerned with MI-based models, such as K-nearest neighbour (KNN), SVD, and neural network-based collaborative filtering (NCF) models. One lakh course review from Coursera are obtained from Kaggle for analysis. Python is used to implement this work. Performance indicators including the hit rate (HR), average reciprocal hit ranking (ARHR), and mean absolute error (MAE) are used to assess how well these models function. From the results, it can be shown that KNN outperforms other models in terms of higher HR and ARHR and lower MAE values [11]. In [12] decision support system is specifically created to be able to offer assistance in selecting graduate-level courses. Python with flask and the HTML programming language were used to create this application. Providing input data in the form of demographic, academic, and personal features is the first step in using this decision support system to choose courses. Pre-processing and data cleaning techniques are applied on input data. Machine learning algorithms such random forests using bagging assembler techniques is applied on processed data result of which developed model which suggest suitable career course for student for mark predictions, compared data mining methods including J-48, Random Forest, Naive Bayes, Naive Bayes Multinomial, K-Star, and IBK. The outcomes show that random forest algorithms outperform other algorithms [13]. In [14] forecast student graduation using artificial neural networks. The suggested system predicts a career option based on the student's psychological state. In [15] developed a methodology to extract highly beneficial association rules from student enrolment data. In [16], to construct a knowledge-based decision approach for picking the best engineering branch, a new attribute selection measurement function (heuristic) was added to the C4.5 DT algorithm. Created an expert system utilizing CLIPS. In [17] to help incoming students select a faculty at Gomal University based on the student's aptitudes, such as intelligence, understanding, comprehension, and mathematical concepts, as well as his or her previous academic success. In [18] the created expert system employs rules to choose the faculty or major that is best for the student based on his skills and aptitudes as shown by the test module results. In [19], students' career choices are influenced by their parents' occupations, the media, and their own inclinations. A proposed system is implemented in [20], and it achieved 94% accuracy, which is better than 5.0 without an adaptive boosting algorithm. This system predicts people's career decisions based on their personality traits, hobbies, and ability to register in courses. In [21] Using ID3, Classification and

Regression Tree, Random Forest, Support Vector Machines, Neural Networks CART, and Multi-Layer Perceptron (MLP), predicted students' expected careers. Using a rule-based knowledge-base developed a web-based Expert System for Career Guidance [22]. In [23] developed a cognitive model that helps students analyze and identify suitable employment opportunities in the software sector. In [24] developed a recommendation system for students to choose their engineering course stream using a feed-forward back propagation probabilistic neural network and classification using fuzzy logic and rough sets. An intuitive career system was created combining binary classification, logistic regression, K-Nearest Neighbours, stochastic gradient descent, and random forest to forecast careers [25]. In [26], an expert system utilizing a machine learning methodology for evaluating students' careers using Artificial Neural Networks was presented. Created an expert system framework for choosing a career utilizing fuzzy logic ID3, J48, and IB1 [27]. In [28] employed Apriori and Nave Bayes method to forecast student's branch prediction, and second is stream analysis of engineering course. A model based on the Naive Bayes algorithm has been developed to predict whether or not a student would enroll in college [29]. For selecting the appropriate field in the CSE/IT stream, [30] designed a recommendation system employing VM, Random Forest decision trees, One Hot encoding, XG enhance Logistic Regression, and Linear Discriminate Analysis. Artificial intelligence algorithms have been developed for the Intelligent Career Counseling Bot. [31]. Framework for Multilevel Expert System to advise Career route Nearest Neighbour (NN) was invented in [32]. Developed a framework for an e-learning course recommendation system using clustering and the Apriori Association Rule Algorithm [33]. While applying for admission to the engineering programme, [34] used the REP Tree method and neural networks to create a model for branch selection. A proposed algorithm that uses Naive Bayes, K-Star, and SVM with the majority voting methodology and an incremental ensemble to reliably forecast a student's career choice [35]. A fuzzy logic-based system for recommending careers to UiTM Perlis students was developed in [36]. Artificial intelligence-based online career counsellor system was developed [37]. The study found that EDM can be used in many different situations, such as identifying at-risk students, prioritizing the educational requirements of various student groups, raising graduation rates, accurately assessing institutional performance, allocating resources on campus, and enhancing topic curriculum renewal [38]. According to [39], IBK displayed high accuracy of 99.680%, while Subset Evaluator performed better than the other three feature selection algorithms. An enhanced Naive Bayesian mining technique was used to obtain the proposed system's accuracy of 86.66% in comparison to using only Naive Bayes [40]. Create the classification model in [41] using Gradient Boosting, Decision Trees, and Support Vector Machines to predict the student's achievement. Researchers find that the Boosting algorithm performs better than alternative classifications. 'Content Similarity Measure (CSM)' algorithm in the healthcare area was proposed in [42, 43, 44]. In [45], predicted student progress in a dispersed setting using a classification technique in data mining.

Section B: Potential Research Gaps Identified

Following is the summarized list of potential research gaps identified through extensive literature from section A

1) The earlier study work does not address the problem of finding impacting variables from all demographics, personal, intellectual, and social components while selecting a course.

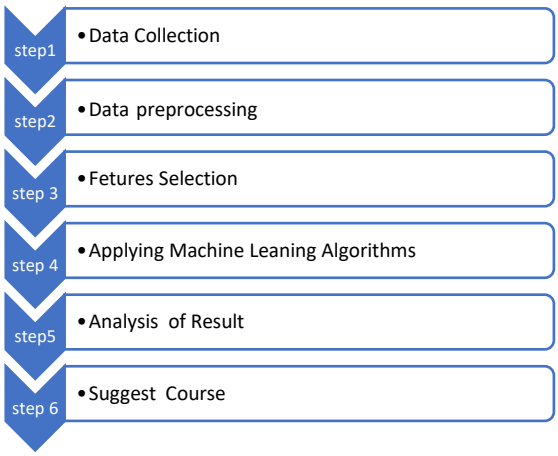
2) The authors discovered that a Proposed Decision support system framework will be helpful for course selection.

METHODOLOGY

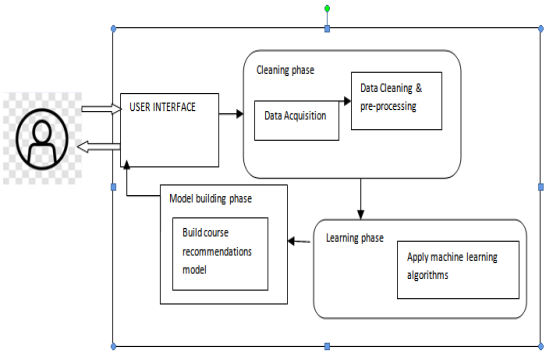
A descriptive research was conducted to design decision support system to predict course using random forest algorithm. The current study employed a survey-based descriptive research methodology. Online surveys were utilized to gather information from students enrolled in courses in a variety of fields, including medicine, engineering, business, and management, animation and cyber security. To choose study participants, the convenience sampling method was used. For the study, 138 students' data from all streams were gathered. A questionnaire was created based on the factors found in the literature and distributed at the convenience of those considering graduation.

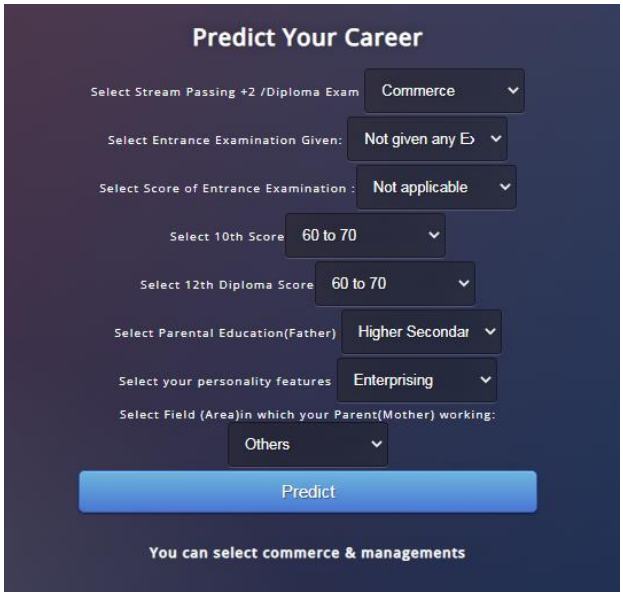
System Concept

This decision support system is specifically created to be able to offer assistance in selecting graduate-level courses. Python with flask and the HTML programming language were used to create this application. Providing input data in the form of demographic, academic, and personal features is the first step in using this decision support system to choose courses. Pre-processing and data cleaning techniques are applied on input data. Machine learning algorithms such random forests using is applied on processed data result of which developed model which suggest suitable career course for student.



Architecture of Proposed DSS





EXPERIMENTAL RESULTS AND DISCUSSION

Data file created from online Google form is transformed to comma separated file to describe the elements influencing students' course selection. The analyses took into account variables like SSC score, HSC board, HSC score, entrance exam, entrance score, birth order, mothers' and fathers' occupations, the number of siblings, time spent with friends, time spent studying, category, parents' occupation field, family income, parent's occupation, area of living, thing like to do most, and tempted variables. In the current work, we used the mutual information gain feature selection method in machine learning. Analysis is based on python code using Google colab. The 10 influencing factors for course selection are shown in Figs. 1.0 and were obtained following feature selection utilizing mutual information gain. According to a study, students primarily take into account 10 features while choosing their courses: SSC Score, Entrance Examination Name, Entrance Examination Score, Category, Mother's Qualification, Father's Job, Personality Trait, and HSC/Diploma Stream. While choosing courses, additional characteristics like birth order, family wealth, location, and HSC score were not taken into account. The accuracy of prediction model utilizing machine learning classifier like random forest was examined in the current study.

Accuracy, precision, recall, and F1-score are used to evaluate the performance of classifier, and these results are visually displayed in Fig. 2.0. The accuracy of machine learning random forest is shown in Fig. 3.0 & Fig. 4.0.

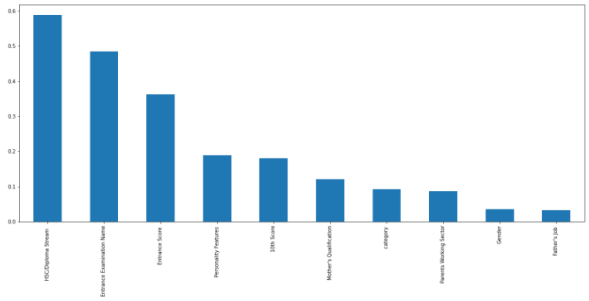


Fig 1.0 Feature Selection-mutual information gain

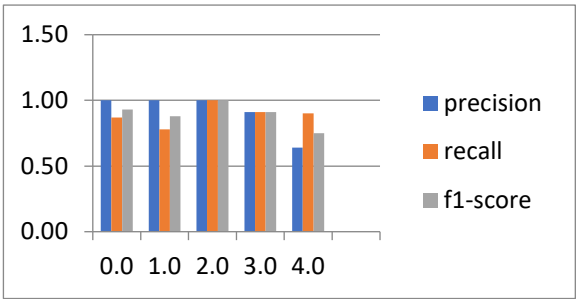


Fig.2.0 precision, recall, F1-score

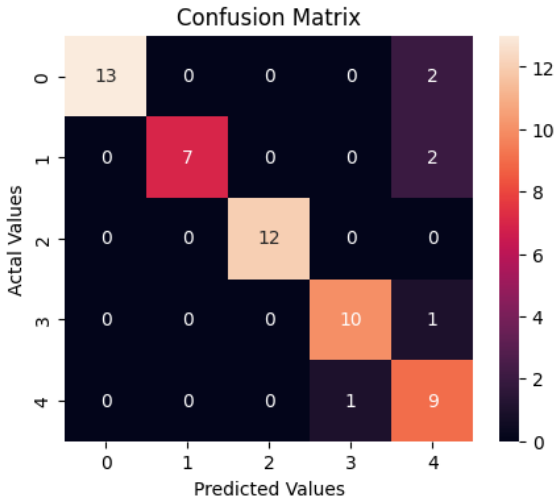


Fig.3.0 Confusion matrix of random forest classifier.

	precision	recall	f1-score	support
0.0	1.00	0.87	0.93	15
1.0	1.00	0.78	0.88	9
2.0	1.00	1.00	1.00	12
3.0	0.91	0.91	0.91	11
4.0	0.64	0.9	0.75	10
accuracy			0.89	57
macro avg	0.91	0.89	0.89	57
weighted avg	0.92	0.89	0.9	57

Fig.4.0 Accuracy report of random forest classifier.

CONCLUSION AND FUTURE ENHANCEMENTS

In this study, survey conducted on the students who are perusing graduation course from Medical, Engineering, Commerce & Management, Animation & Cyber Security streams has aimed to identify influencing factors of students while course choice decision. Machine learning features selection methods such as Mutual information gain was applied on the collected data to find out the important features which were considered by the students while course selection. According to the analysis 10 factors namely SSC score, Entrance Examination name, Entrance score, personality, Father's occupation, mother's qualification, hobby, category, HSC/Diploma stream are mostly considered by students while course selection. while other factors such family income, area of living, fathers' qualification, HSC board, time spend for study, time set with friends, number of siblings birth order, family income were not considered while course selection. The experimental results show the performance accuracy of random forest classifier is 89%.

Research results will help students and parents in decision-making process while course selection.

In the future, the author wants to compare the accuracy of models by applying other Ensemble classifiers.

REFERENCES

[1]. Sohaib Latif, Fang XianWen , and Li-li Wang," Intelligent decision support system approach for predicting the performance of students based on three-level machine learning technique", Journal of Intelligent Systems 2021; 30: 739–749, https://doi.org/10.1515/jisys-2020-0065 received June 23, 2020; accepted March 19, 2021

[2]. Made Sudarma1, Anak Agung Kompiang Oka Sudana2, Irwansyah Cahya," Decision Support System for the Selection of Courses in the Higher Education using the Method of Elimination Et Choix Tranduit La Realite",

International Journal of Electrical and Computer Engineering (IJECE) Vol. 5, No. 1, February 2015, pp. 129~135 ISSN: 2088-8708

[3]. Ahmad A. Kardan, Hamid Sadeghi," A Decision Support System for Course Offering in Online Higher Education Institutes", International Journal of Computational Intelligence Systems, Vol. 6, No. 5 (September 2013), 928-942

[4] zameer gulzar,aany leema,Gerard Deepak"PCRC:Personalized course recommender system based on hybrid approach ", 1877-0509 © 2018 The Authors. Published by Elsevier B.V. Peer-review under responsibility of the scientific committee of the 6th International Conference on Smart Computing and Communications 10.1016/j.procs.2017.12.067

[5]. Svetlana Mansmann and Marc H. Scholl" Decision Support System for Managing Educational Capacity Utilization", Erschienen in: IEEE Transactions on Education; 50 (2007), 2. - S. 143-150 https://dx.doi.org/10.1109/TE.2007.893175

[6]. ASMAE BENTALEB1, JAAFAR ABOUCHABAKA1 "ENSEMBLE LEARNING FOR MINING EDUCATIONAL DATA" Journal of Theoretical and Applied Information Technology 15th May 2022. Vol.100. No 9

[7] M. Ramaswami and R. Bhaskaran(2009)" A Study on Feature Selection Techniques in Educational Data Mining" JOURNAL OF COMPUTING, VOLUME 1, ISSUE 1, DECEMBER 2009, ISSN: 2151-9617 HTTPS://SITES.GOOGLE.COM/SITE/JOURNALOFCOMPUTING/

[8]. Umesh Kumar Sah1, Mrs. Awantika Singh2," Student Career Prediction Using Machine Learning", International Journal of Scientific Development and Research (IJSDR) www.ijsdr.org, May 2022 IJSDR | Volume 7 Issue 5 ISSN: 2455-2631

[9] Chao-Yen Wu, ~'~'anciaco Irazusta, and] ames T. Lancaster" A DECISION SUPPORT SYSTEM FOR COLLEGE SELECTION", Proceedings of the 14th Annual Conference on Computers and Industrial Engineering

[10]. Ditika Bhanushali, Seher Khan, Mohommad Madhia, Shoumik Majumdar (2018)," Student Performance Prediction And Analysis", International Journal of Advanced Research in Computer and Communication Engineering,ISO 3297:2007 Certified Vol. 7, Issue 3, March 2018

[11]. Kalyan Kumar Jena 1, Sourav Kumar Bhoi 1, Tushar Kanta Malik 1, Kshira Sagar Sahoo 2,3, N Z Jhanjhi 4, *, Sajal Bhatia 5 and Fathi Amsaad 6"E-Learning Course Recommender System Using Collaborative Filtering Models" Jena, K.K.; Bhoi, S.K.; Malik, T.K.; Sahoo, K.S.; Jhanjhi, N.Z.; Bhatia, S.; Amsaad, F. E-Learning Course Recommender System Using Collaborative Filtering Models. Electronics 2023, 12, 157. https:// doi.org/10.3390/electronics12010157

[12] Cut Fiarni1) *, Evasaria M. Sipayung2), Prischilia B.T. Tumundo3)," Academic Decision Support System for Choosing Information Systems Sub Majors Programs using Decision Tree Algorithm", Journal of Information Systems Engineering and Business Intelligence, Vol.5, No.1, April 2019 Available online at: http://e-journal.unair.ac.id/index.php/JISEBI.

[13] Bhrigu Kapur, Nakin Ahluwalia," Comparative Study on Marks Prediction using Data Mining and Classification Algorithms" International Journal of Information Systems in the Service Sector,Volume 9 • Issue 1 • January-March 2017

[14] Asep Nurhuda, Dewi Rosita (2017),” Prediction Student Graduation on Time Using Artificial Neural Network on Data Mining Students STMIK Widya Cipta Dharma Samarinda”, ICEEG '17, June 14-16, 2017, Turku, Finland. Copyright 2017 ACM 978-1-4503-5248-2/17/06…$15.00., DOI: http://dx.doi.org/10.1145/3108421.3108431

[15] Zailani Abdullaha,*, Tutut Herawanb, Noraziah Ahmadb, Mustafa Mat Derisc “Mining significant association rules from educational data using critical relative support approach”, Procedia - Social and Behavioral Sciences 28 (2011) 97 – 101

[16] Malaya Dutta Borah, Rajni Jindal, Daya Gupta Ganesh Chandra Deka (2011) “Application of knowledge-based decision technique to Predict student enrollment decision”, 2011 International Conference on Recent Trends in Information Systems, 978-1-4577-0792-6/11/$26.00 ©2011 IEEE

[17]. Muhammad Zaheer Aslam*, Nasimullah, Abdur Rashid Khan A Proposed Decision Support System/Expert System for Guiding Fresh Students in Selecting a Faculty in Gomal University, Pakistan, Industrial Engineering Letters www.iiste.org ISSN 2224-6096 (Print) ISSN 2225-0581(Online) Vol 1, No.4, 2011

[18]. Subhalaxmi Panda, P. A Pattanaik , Tripti Swarnkar (2017)” A Higher Education Predictive Model Using Data Mining Techniques”, individual papers by the papers’ authors

[19] Esra Mahsereci Karabuluta*, Selma Ayşe Özelb, Turgay İbrikçib,c,” A comparative study on the effect of feature selection on classification accuracy” 2212-0173 © 2012 Published by Elsevier Ltd. doi: 10.1016/j.protcy.2012.02.068

[20] Nikita Gorad, Ishani Zalte, Aishwarya Nandi, Deepali Nayak (2017)” Career Counselling Using Data Mining”, International Journal of Innovative Research in Computer and Communication Engineering (An ISO 3297: 2007 Certified Organization) Vol. 5, Issue 4, April 2017, ISSN(Online): 2320-9801 ISSN (Print): 2320-9798

[21]. Md. Yeasin Arafath, Mohd. Saifuzzaman(2018),”Predicting Career Using Data Mining”, 2018 International Conference on Computing, Power and Communication Technologies (GUCON) Galgotias University, Greater Noida, UP, India. Sep 28-29, 2018

[22]. S. Saraswathi1, M. Hemanth Kumar Reddy2, S. Udaya Kumar3, M. Suraj4, Sk. Khaja Shafi5,” DESIGN OF AN ONLINE EXPERT SYSTEM FOR CAREER GUIDANCE”, Volume: 03 Special Issue: 07 | May-2014, Available @ http://www.ijret.org, IJRET: International Journal of Research in Engineering and Technology eISSN: 2319-1163 | pISSN: 2321-730

[23]. V. Chandra Prakash1, J.K. RSastry2*G. Reeshmika3, M.Pavani 4, P.Chikitha Sree5 , J.S Ravi Teja6, Development of a Comprehensive and Integrated Expert System for Career Assessment based on Cognitive models International Journal of Emerging Trends in Engineering Research, 7(11), November 2019, 617- 627

[24]. Grewal DS and Kaur K (2016) “Developing an Intelligent Recommendation System for Course Selection by Students for Graduate Courses”, Business and Economics Journal, Bus Eco J 2016, 7:2DOI: 10.4172/2151-6219.100020

[25]. Rucha Hemant Rangnekar,Khyati Pradeep Suratwala,Sanjana Krishna,Dr.

Sudhir Dhage(2018),"Career PredicData Mining And Linear Classification ",978-1-5386-5257-2/18/$31.00 2018 IEEE

[26]. V. Sai kuswanth, Gogineni Krishna Chaitanya, B.Sekhar Babu, Uppuluri Lakshmi Soundharya (2019),"Machine Learning Approach for Student Career assessment in the modern world",International Journal of Recent Technology and Engineering (IJRTE) ISSN: 2277-3878, Volume-7, Issue-6S, March 2019 699

[27]. M.L.Waghmode, Jamsandekar(2016)," Expert system for Career Selection: A Classifier Model", Volume 4, Issue 1, January 2016 International Journal of Advance Research in Computer Science and Management Studies , ISSN: 2321-7782 ,Research Article / Survey Paper / Case Study Available online at: www.ijarcsms.com

[28]. Heena Sabnani, Mayur More, Prashant Kudale, Prof. Surekha Janrao (2018)" Prediction of Student Enrolment Using Data Mining Techniques", International Research Journal of Engineering and Technology (IRJET) e-ISSN: 2395-0056 Volume: 05 Issue: 04 | Apr-2018 www.irjet.net p-ISSN: 2395-0072.

[29]. A Framework For Analytical Services Using Data Mining Techniques to Predict the Enrollment of Student At A University – A Case Study.

[30]. lakshmi prasanna, DR.D.Haritha (2019)"Smart Career Guidance and Recommendation System", © IJEDR 2019 | Volume 7, Issue 3 | ISSN: 2321-9939

[31]. Dr. Gufran Ahmad Ansari (2017) "Career Guidance throughMultilevel Expert System Using Data Mining Technique" I.J. Information Technology and Computer Science, 2017, 8, 22-29 Published Online August 2017 in MECS DOI: 10.5815/ijitcs.2017.08.03

[32]. Sunita B. Aher (2014)," EM&AA: An Algorithm for Predicting the Course Selection by Student in e-Learning Using Data Mining Techniques", J. Inst. Eng. India Ser. B (January–March 2014) 95(1):43–54, DOI 10.1007/s40031-014-0074-3

[33]. Heena Sabnani, Mayur More, Prashant Kudale, Prof. Surekha Janrao (2018)" Prediction of Student Enrolment Using Data Mining Techniques", International Research Journal of Engineering and Technology (IRJET) e-ISSN: 2395-0056 Volume: 05 Issue: 04 | Apr-2018 www.irjet.net p-ISSN: 2395-0072

[34]. Ms. Roshani Ade, Dr. P. R. Deshmukh (2014)," An incremental ensemble of classifiers as a technique for prediction of student's career choice", 978-1-4799-3486-7/14/$31.00_c 2014 IEEE

[35]. Career Path R Tajul Rosli Razak1, Muhamad Arif Hashim2, Noorfaizalfarid Mohd Noor3, Iman Hazwam Abd Halim4, Nur Fatin Farihin Shamsul recommendation System for UiTM Perlis Students Using Fuzzy Logic, 978-1-4799-4653-2/14/$31.00 © 2014 IEEE

[36] Kartikey Joshi, Amit Kumar Goel, Tapas Kumar 2020, Online Career Counsellor System based on Artificial Intelligence: An approach IEEE 7th International Conference on Smart Structures and Systems ICSSS 2020

[37]. Abdulmohsen Algarni (2016) "Data Mining in Education",(IJACSA) International Journal of Advanced Computer Science and Applications,Vol. 7, No. 6, 2016456 | P a g e www.ijacsa.thesai.org.

[38]. T.Velmurugan, C. Anuradha (2016) "Performance Evaluation of Feature Selection Algorithms in Educational Data Mining", International Journal of Data Mining Techniques and Applications Volume: 05 Issue: 02 December 2016

Page No.131-139 ISSN: 2278-2419

[39]. Shumba, A. & Naong, M. (2012). Factors Influencing Students' Career Choice and Aspirations in South Africa. Central University of Technology, Free State, 169-178.Retrieved from http://www.ashumba@cut.ac.za&almondshumba@yahoo.com

[40]. Khoo, K., Ban, T.K., Neng, C.Y., Hooi, B.K. & Joan, C.Y. (2015). Students Choices of choosing Colleges and Course of Study in Penang. ABC Research Alert, 3(1), 1-7. Retrieved from http://www.abcreal.weebly.com

[42]. Yashoda Barve, Jatinderkumar R. Saini “, Detecting and classifying online health misinformation with ‘Content Similarity Measure (CSM)’ algorithm: an automated fact-checking-based approach”, The Journal of Supercomputing, Online ISSN: 1573-0484, 2023

[43] Yashoda Barve, Jatinderkumar R. Saini “, Detecting and classifying online health misinformation with ‘Content Similarity Measure (CSM)’ algorithm: an automated fact-checking-based approach”, The Journal of Supercomputing, Online ISSN: 1573-0484, 2023

[44] Yashoda Barve, Jatinderkumar R. Saini Kaushika Pal3, Ketan Kotecha” A Novel Evolving Sentimental Bag-of-Words Approach for Feature Extraction to Detect Misinformation”, (IJACSA) International Journal of Advanced Computer Science and Applications, Vol. 13, No. 4, 2022

[45] Krina Parmar, Prof. Dineshkumar Vaghela, Dr Priyanka Sharma (2015),” PERFORMANCE PREDICTION OF STUDENTS USINGDISTRIBUTED DATA MINING”, IEEE Sponsored 2nd